Module-2

Banking Operations & Digital Banking

Core Banking Solutions

- The **core banking solutions** refer to a **centralized banking system** that allows banks **to deliver services to customers** in real time, across all branches and digital channels, **through a single integrated platform**.
- Customers in a CBS environment are **bank-centric rather than branch-centric**, i.e. they can access their accounts from **any branch or digital channel**.

❑ Key Features of Core Banking Solutions:

- **Centralized Database**
  - All customer and transactional data are stored in a **central server**.
  - Ensures consistency and real-time updates.
- **Anywhere Banking**
  - Customers can deposit, withdraw or transfer funds from **any branch.**
  - Enables ATM, internet banking and mobile banking.
- **Real-Time Processing**
  - Transactions are updated instantly.
  - Reduces delays and reconciliation errors.
- **Multi-Channel Integration**
  - CBS integrates channels like ATMs, UPI, debit/credit cards, internet banking, mobile apps.
  - Supports omni-channel banking experience.
- **Scalability and Security**
  - Handles large volumes of transactions efficiently.
  - Includes encryption, authentication and audit controls.

❑ Benefits of Core Banking Solutions:

- **For Banks:**

- Improved operational efficiency
- Centralized control and monitoring
- Faster product launch
- Better risk and compliance management

- **For Customers:**

- Convenience and faster services
- Access to digital banking channels
- Uniform service quality across branches

❑ Challenges of Core Banking Solutions:

- High implementation and migration costs.
- Cyber-security risks due to centralized data storage.
- Downtime during system upgrades.
- Dependence on vendors for technical support.

Traditional Banking *vs* Core Banking Solutions (CBS)

Aspect	Traditional Banking	Core Banking Solutions (CBS)
System Type	Decentralized & branch-centric	Centralized & customer-centric
Customer Access	Customer tied to a specific branch	Customer can access from any branch
Transaction Processing	Batch processing, delayed updates	Real-time processing
Data Storage	Separate databases for each branch	Single centralized database
Service Speed	Slow and manual	Fast, real-time and automated
Digital Services	Limited or unavailable	Fully supports ATM, UPI, internet & mobile banking
Reporting & MIS	Manual, time-consuming	Automated, real-time reports
Customer Experience	Inconvenient and inconsistent	Seamless and uniform
Scalability	Difficult to scale	Easily scalable
Security & Compliance	Limited monitoring	Strong audit trails and controls

Integrating AI with Banking

- Core Banking Solutions provide **structured & centralized data**, which serves as a strong foundation for incorporating **AI and Data Analytics** in banking.
- AI systems are **layered on top of CBS** using APIs and data pipelines to analyze transactions, customer behavior and operational data.
- Basically, its use is an extension to the conventional services of CBS.
- CBS modernize banking operations by enabling centralized and real-time services. When integrated with AI, CBS transforms banks into **intelligent, data-driven institutions** capable of delivering personalized services, improving risk control and enhancing operational efficiency.

Applications of AI Integrated with CBS

a) Fraud Detection:

- AI analyzes transaction patterns in real time.
- Detects anomalies such as unusual spending or location changes.
- Reduces fraud and false alerts.

b) Credit Scoring & Risk Assessment:

- AI uses CBS data (such as transaction history, repayment behavior).
- Enables faster and more accurate loan decisions.

c) Personalized Banking:

- AI recommends products based on customer spending patterns.
- Improves cross-selling and customer engagement.

d) Chatbots & Virtual Assistants:

- AI chatbots access CBS data to provide:
  - Account balance
  - Transaction history
  - Loan details
- Ensures 24×7 customer support.

e) Predictive Analytics:

- Forecasts any potential customer churning.
- Predicts loan defaults and confirms the cash flow trends.

Benefits & Challenges

- **Benefits of CBS–AI Integration:**
- Faster decision-making
- Enhanced customer experience
- Improved risk management
- Cost reduction through automation
- Data-driven business strategies

- **Challenges of CBS–AI Integration:**
- Data privacy and security risks
- High implementation cost
- Need for skilled workforce
- Integration with legacy CBS platforms

India-Specific Banks: CBS and AI Integration

- State Bank of India (SBI):

- **AI-based fraud detection** for UPI and card transactions.
- **Chatbot "SIA"** (SBI Intelligent Assistant) accesses CBS data for customer queries.

- HDFC Bank:

- **EVA chatbot** provides account and product information using CBS data.
- **AI-driven credit analytics** for retail and MSME lending.
- **Transaction monitoring systems** for fraud detection.

- ICICI Bank:

- **AI-powered chatbots** for retail and corporate customers.
- **Robotic Process Automation (RPA)** integrated with CBS for KYC and account opening. (automated & faster back-office processes)
- **Predictive analytics** for loan default risk.

E-banking tools: NEFT, RTGS, IMPS, Mobile banking

- **E-banking tools** are electronic payment and service channels that enable customers to perform banking transactions **digitally, securely and efficiently.**

1. NEFT (National Electronic Funds Transfer)

- NEFT is a **nationwide electronic payment system** that allows **one-to-one transfer of funds** between bank accounts in India.
- It **operates 24×7** and settlements are done in **half-hourly batches.**
- No minimum or maximum transfer limit (subject to bank policy).
- Requires **beneficiary details** (account number, IFSC).
- Salary payments, Utility bill payments, Vendor payments, etc.

2. RTGS (Real Time Gross Settlement)

- RTGS is a system where **funds are transferred individually and in real time**, without any batching.
- It operates **24×7** and the **minimum amount is ₹2 lakh.** No upper limit.
- It is used for **high-value transactions** which must reflect immediately.
- Property transactions, Large corporate payments, Inter-bank settlements.

3. IMPS (Immediate Payment Service)

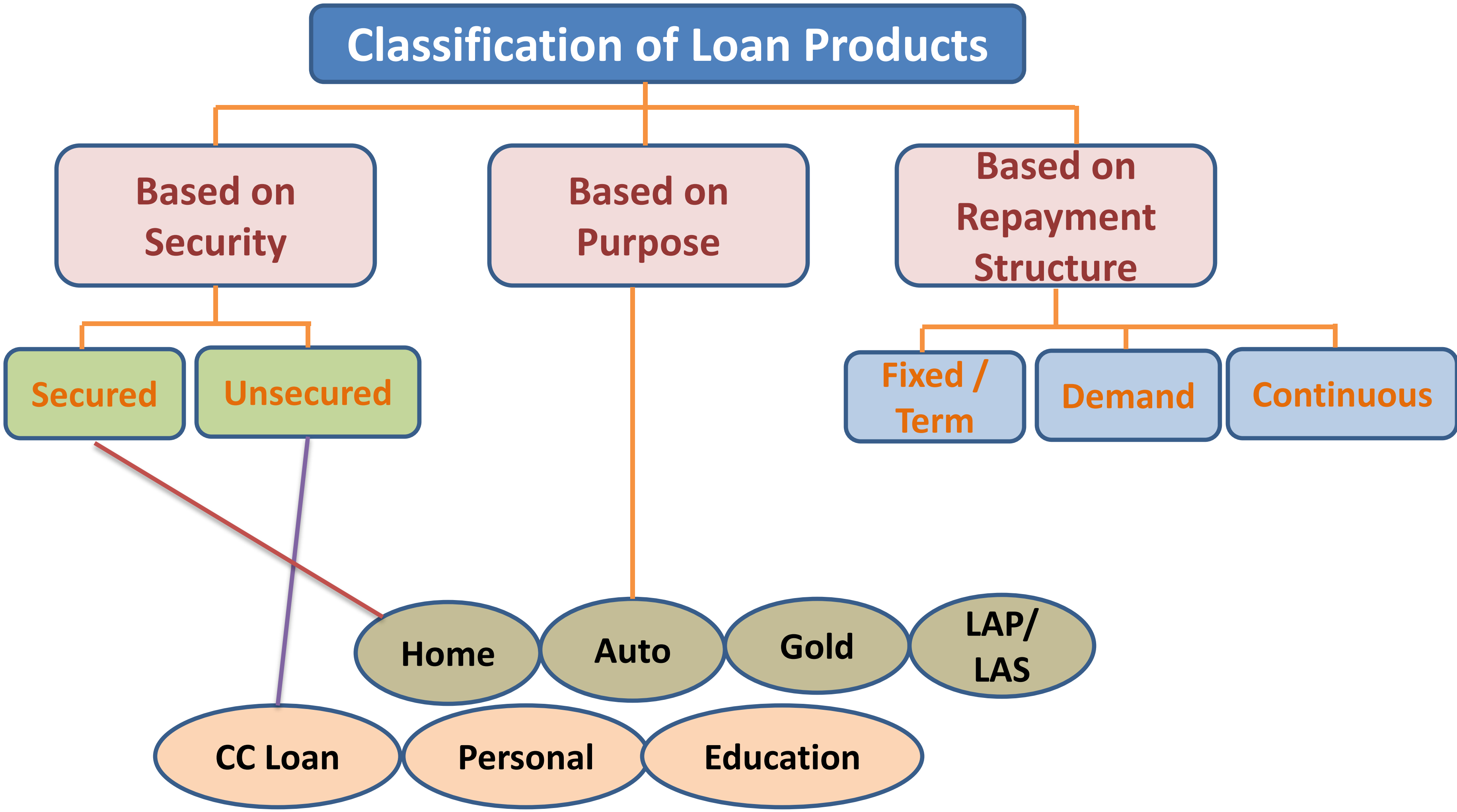
- IMPS enables **instant, real-time fund transfers** through mobile phones and internet banking.
- Available **24×7, including holidays**.
- Instant credit to beneficiary.
- Transaction limit decided by banks (generally between **₹2–5 lakh**), with **charges from NIL to ₹25 + GST** per transaction.
- Emergency fund transfers, urgent Person-to-person payments.

4. Mobile Banking

- Mobile banking allows customers to perform banking transactions using **Apps provided by banks**.
- Key functionalities include:
  - Balance enquiry and statement generation.
  - Fund transfers (NEFT, RTGS, IMPS, UPI).
  - Bill payments and recharges.
  - Loan and FD related services.
  - Secure authentication (PIN, OTP, biometrics)
- Eg. SBI YONO, HDFC Mobile Banking, ICICI iMobile Pay, Axis Mobile.

Feature	NEFT	RTGS	IMPS	Mobile Banking
Settlement	Batch	Real-time	Real-time	Real-time
Availability	24×7	24×7	24×7	24×7
Minimum Amount	No minimum	₹2 lakh	Bank-defined	No minimum
Speed	Moderate	Immediate	Instant	Instant
Transaction Type	Low–medium value	High value	Instant P2P	All services

Loan Products



- **Secured Loans (Backed by Assets):**

- Backed by collateral, lower interest rate and lower credit risk for banks.

- **Home Loan:** For purchasing or constructing property.
- **Loan Against Property (LAP):** Mortgage loan using property as collateral.
- **Gold Loan:** Funds secured by gold jewelry.
- **Vehicle/Auto Loan:** Specifically for buying cars or two-wheelers.
- **Loan Against Securities:** Borrowing against shares, mutual funds, or insurance policies.
- **Machinery/Equipment Loan:** For industrial equipment purchase.

- **Unsecured Loans:**

- No collateral is required, higher interest rate and sanctioned by banks based on creditworthiness.

- **Personal Loan:** Used for various personal, medical, or travel needs.
- **Credit Card Loans:** Instant, high-interest loans against a credit card limit.
- **Education Loan:** Specifically for higher studies.
- **Short-term Business Loan:** Immediate cash for business needs.

Business & Corporate Loans

1. MSME / Business Loans

- Working capital and business expansion.
- Examples:
  - Term loans
  - Working capital loans
  - Mudra loans (Shishu, Kishor, Tarun)

2. Working Capital Finance

- For day-to-day operations, short-term in nature.
- Types:
  - Cash Credit (CC)
  - Overdraft (OD)
  - Bill discounting

3. Term Loans

- It is used for capital expenditure
- It has fixed repayment schedule
- Medium or long-term
- Common in industrial and infrastructure projects.

- **Specialized Loan Products**

1. Agricultural Loans

- For crop production, farm equipment, often subsidized
- Examples:
  - Crop loans
  - Kisan Credit Card (KCC)

2. Gold Loans

- Secured by gold ornaments
- Quick processing
- Short tenure

3. Loan Against Securities

- Secured by shares, mutual funds, insurance policies, etc.
- Used for liquidity without selling investments.

Know Your Customer (KYC)

- KYC is a **mandatory process** followed by banks and financial institutions to **verify the identity and address of customers** before providing any financial services.
- KYC helps banks understand **who the customer is**, **the nature of their business** and **the risk involved**.

- **Why was KYC required in Banking?**

- KYC was introduced to **protect the banking system and economy** from misuse.

- Avoid Identity Fraud: Ensures accounts are opened by genuine, existing individuals.
- Prevent Money Laundering: Stops illegal money from entering the financial system.
- Curb Terrorist Financing: Prevents funding of unlawful and terrorist activities.
- Improve Financial Transparency: Helps track customer deposits & withdrawals and detect suspicious transactions.

KYC Norms

- KYC norms in India are issued and regulated by the **Reserve Bank of India (RBI)** under **AML (Anti-Money Laundering) guidelines**.
- KYC norms ensure customer identification, prevent financial crimes and maintain the integrity of the banking system.
- Valid Documents are:

Purpose	Documents
Proof of Identity	Aadhaar, Passport, Voter ID, PAN, Driving Licence
Proof of Address	Aadhaar, Passport, Utility bill, Bank statement
PAN	Mandatory for most financial transactions/services

a) Full KYC

- Complete verification using all valid documents
- No transaction limits

b) Simplified / Minimum KYC

- Limited documentation
- Transaction limits apply

c) e-KYC

- Aadhaar-based biometric or OTP verification
- Faster and paperless

d) Video KYC

- Live video interaction with customer
- Accepted by RBI for remote on-boarding

KYC Risk Management

- KYC Risk Management refers to the process of **identifying, assessing, monitoring and mitigating the risks associated with customers** to prevent **money laundering, terrorist financing, fraud and regulatory non-compliance**.
- Banks classify customers into risk categories:

Risk Level	Characteristics	KYC updated
Low Risk	Salaried employees, pensioners	Every 2 years
Medium Risk	Self-employed professionals	Every 5 years
High Risk	PEPs, NRIs, NGOs, trusts	Every 10 years

Types of Risks in KYC that require CDD/EDD:

A. Customer Risk

- Nature of customer (individual, company, trust)
- Politically Exposed Persons (PEPs)
- Non-resident customers

B. Geographic Risk

- Customers from high-risk or sanctioned countries (Iran, Myanmar, North Korea, etc.)
- Regions with weak AML controls

C. Product & Service Risk

- Cash-intensive businesses accounts (restaurants, retailers, etc.)
- Cross-border services (remittances P2P, C2B e-commerce, etc.)

D. Transaction Risk

- Unusual transaction patterns
- Large cash deposits

Quiz on KYC